
Subject: Digital Technologies
Class Level: Senior Secondary School 2 (SS2)
Academic Year: Full Session
Duration per Week: 2–3 periods

 First Term Scheme of Work (SS2)

Week 1: Revision of SS1 Work

- Recap: Networks, web dev basics, databases, cybersecurity.

Week 2: Computer Architecture (Advanced)

- CPU design, memory hierarchy, buses, and data flow.

Week 3: Operating Systems (File & Process Management)

- Concepts: multitasking, scheduling, file permissions.

Week 4: Data Structures & Algorithms (Intro)

- Arrays, lists, flowcharts, pseudocode.

Week 5: Advanced Word Processing & Desktop Publishing

- Page layout, newsletters, brochures.

Week 6: Spreadsheet Data Modelling

- Scenario analysis, pivot tables, dashboards.

Week 7: Web Development (CSS & JavaScript Basics)

- Styling with CSS; introduction to interactivity with JS.

Week 8: Database Management Systems (Intermediate)

- Relational DBMS, SQL basics (SELECT, INSERT, UPDATE).

Week 9: Python Programming (Intermediate)

- Functions, error handling, simple projects.

Week 10–11: ICT & Society: Ethical and Legal Issues

- Digital law, copyright, intellectual property, data protection.

Week 12: Revision & Examination

Second Term Scheme of Work (SS2)

Week 1: Revision of First Term

Week 2: Networking Protocols

- TCP/IP, HTTP, FTP, SMTP, security protocols.

Week 3: Cloud Computing (Detailed Study)

- SaaS, PaaS, IaaS, cloud storage examples.

Week 4: Advanced Spreadsheet Analysis

- Macros, automation, financial modelling.

Week 5: Database Queries & Reports

- SQL queries with conditions, grouping, reporting.

Week 6: Web Development (Forms & Interactivity)

- HTML forms, CSS styling, basic JavaScript validation.

Week 7: Python Projects (Problem Solving)

- Build calculators, quizzes, or record systems.

Week 8: Multimedia & Animation Tools

- Creating animations and short interactive videos.

Week 9: Cybersecurity (Ethical Hacking Basics)

- Password security, penetration testing concepts.

Week 10–12: Project Work + Examination

Third Term Scheme of Work (SS2)

Week 1: Revision of Second Term

Week 2: Artificial Intelligence (Detailed Concepts)

- AI in daily life; machine learning basics.

Week 3: Robotics (Practical Concepts)

- Sensors, actuators, simple robotic programming.

Week 4: Internet of Things (IoT)

- Smart homes, smart agriculture, wearable devices.

Week 5: Data Science & Big Data

- Introduction to large datasets, visualization.

Week 6: Mobile Application Development (Intermediate)

- App design, testing, and publishing basics.

Week 7: ICT and Entrepreneurship (Startups & Innovation)

- Tech business models, freelancing, e-commerce.

Week 8: ICT for Sustainable Development

- Role in education, health, environment.

Week 9: Capstone Project (Web/App Development)

- Students develop an ICT-based solution to real-life problems.

Week 10: General Revision & Examination